

Project Presentation of Industry Oriented Hands on Experience (22CS422)

On

Sitekraft — AI-Powered Website Builder

Koustubh Kukreti

2210990517

Sneha Chaudhary

2210990857

Supervised By

Dr. Monika Aggarwal

Department of Computer Science and Engineering,
Chitkara University, Punjab

- Modern businesses require a strong online presence.
- Traditional website development requires technical expertise.
- Existing builders rely heavily on static templates.
- Sitekraft uses AI to generate complete React websites automatically.
- The platform integrates code generation, copywriting, image integration, and deployment.

- Website development is expensive and time-consuming.
- Non-technical users struggle with frontend, backend, and deployment workflows.
- Drag-and-drop builders provide limited flexibility.
- Small businesses need faster and smarter website creation solutions.

Solution:

- An autonomous AI-powered website generation platform.

Project Objectives

- Generate production-ready React websites using AI
- Automate cloud deployment workflows
- Integrate AI-generated website content
- Support multiple business categories
- Reduce technical barriers for users
- Generate responsive modern UI automatically

Workflow:

- User enters business information
- React frontend sends data to FastAPI backend
- Gemini AI generates React components and content
- Image APIs fetch contextual images
- Website is deployed automatically using Vercel

Technologies:

- Frontend: React + Vite
- Backend: FastAPI
- AI: Gemini AI
- Deployment: Vercel

Key Features

- **Conversational UI** — chat with the AI agent to describe your website
- **Full React/Vite generation** — produces real component-based code, not templates
- **Auto-deployment** — deploys generated sites directly to Vercel
- **Stock image integration** — pulls relevant images from Pexels & Pixabay
- **Supports** e-commerce, booking pages, media uploads, and contact forms

Frontend

- Multi-step onboarding workflow
- Dynamic preview system
- Responsive UI rendering

Backend

- FastAPI orchestration system
- API handling and deployment automation
- AI prompt management

AI System

- JSX component generation
- Layout generation
- Marketing content generation

- Sitekraft automates website development workflows using AI.
- The system combines AI-generated code, deployment automation, and responsive UI generation.
- The platform reduces technical barriers for businesses and non-technical users.
- The project demonstrates practical application of AI in software engineering.
- Final Statement:
- “AI can transform website development into an autonomous workflow.”

- Multi-page website generation
- SEO optimization
- Custom domain integration
- Real-time editing support
- Mobile application support
- Analytics integration
- AI streaming generation



Thank You